

## **Group Work Instructions**

### **Objective**

The goal of this project is to design and implement a relational database system that addresses a real-world problem aligned with a Sustainable Development Goal (SDG). The project involves the creation of an Entity-Relationship Diagram (ERD), a detailed database schema, SQL code implementation, and the use of GitHub for collaboration and version control.

### **Project Components**

1. Project Idea Definition
  - ✓ Detailed Explanation

Start by clearly defining your chosen project topic. Provide a detailed description of the problem your database system is intended to solve. Explain how it aligns with the SDG you are focusing on and its relevance in the Kenyan context.

- ✓ Scope and Objectives

Outline the scope of your project, including specific objectives you aim to achieve with the database system. Describe the key functionalities and features that the database should support.

- ✓ Stakeholders

Identify the main stakeholders who will interact with or benefit from your database system. Explain how your project will address their needs.

2. Entity-Relationship Diagram (ERD)
  - ✓ Design

Develop an ERD to visually represent the entities, attributes, and relationships within your database system. The ERD should clearly define the structure of your database, showing how data will be organized and related.

- ✓ Entities and Attributes

Identify and define the entities involved in your project. List their attributes and indicate primary and foreign keys where applicable.

- ✓ Relationships

Clearly define the relationships between entities, including cardinality and participation constraints.

3. Database Schema
  - ✓ Normalization

Based on your ERD, design a normalized database schema. Ensure that your schema is free of redundancy and anomalies.

- ✓ Tables and Fields

Translate your ERD into a set of database tables. Define the fields (columns) for each table, specifying data types and constraints (e.g., NOT NULL, UNIQUE, etc.).

- ✓ Indexes

Consider indexing strategies to improve the performance of your database for key queries.

#### 4. SQL Code Implementation

- ✓ Database Creation

Write SQL scripts to create the database and tables according to your schema. Ensure that all necessary constraints (e.g., primary keys, foreign keys, unique constraints) are implemented.

- ✓ Data Manipulation

Provide SQL code for basic CRUD (Create, Read, Update, Delete) operations. Include example queries that demonstrate how data can be inserted, retrieved, updated, and deleted from your database.

- ✓ Advanced Queries

Develop SQL queries to address specific use cases relevant to your project. This could include joins, subqueries, aggregate functions, and complex data retrieval based on the needs of your stakeholders.

#### 5. GitHub Collaboration

- ✓ Repository Setup

Create a GitHub repository for your project. This repository should be used to track all changes and collaborate with your group members. Ensure that your repository is well-organized with clear folder structures and naming conventions.

- ✓ Version Control

Use Git for version control, ensuring that all changes to the project files (including ERD diagrams, SQL scripts, and documentation) are committed with descriptive messages.

- ✓ Collaboration

Each group member should contribute to the project by working on different aspects of the database design and implementation. Use branches and pull requests to manage changes and ensure quality control.

#### 6. Final Presentation and Report

- ✓ Project Report

Compile a detailed report that includes your project idea definition, ERD, database schema and GitHub repository link. The report should explain the design decisions made and how the database system meets the project's objectives.

### Submission Guidelines

#### 1. GitHub Repository

Ensure that your final project files are all pushed to the GitHub repository. Share the link to your repository as part of your submission.

## **2. Report**

Submit the project report as a PDF document via the course's submission platform.

### **Evaluation Criteria**

1. Project Definition and Relevance - Clarity and relevance of the project idea in relation to the SDG.
2. ERD and Database Schema - Accuracy, completeness, and normalization of the database design.
3. SQL Implementation - Correctness and efficiency of the SQL code, including advanced query handling.
4. GitHub Use - Effective use of GitHub for collaboration, including proper version control and contribution by all group members.
5. Final Report and Presentation - Quality and clarity of the final report and the ability to communicate the project's outcomes in the report

### 1. Database for Disaster Response Coordination

Design and implement a database to manage data during natural disasters, enabling efficient resource allocation and communication between agencies.

### 2. Accessible Public Transportation System Database

Develop a database to support the design of an inclusive public transportation system, addressing accessibility challenges for people with disabilities.

### 3. Circular Economy Database for E-Waste Management

Build a database to manage the lifecycle of electronic products, focusing on reducing e-waste through recycling, refurbishing, and responsible disposal.

### 4. Urban Green Spaces Database

Create a database to manage urban green spaces, tracking maintenance, community use, and environmental impact.

### 5. Affordable Housing Database

Create a database to track affordable housing projects, including occupancy rates and community development impact.

### 6. Community Health and Nutrition Database

Design a database to manage data on community health initiatives, particularly those addressing malnutrition and access to healthy food options.

### 7. Water Access Management Database

Design a database to track and manage community water points in rural areas, including usage, maintenance schedules, and water quality monitoring.

### 8. School Feeding Program Database

Create a database to manage and monitor school feeding programs, tracking food distribution, student attendance, and nutritional impact.

### 9. Affordable Housing Allocation System

Develop a database to manage applications for affordable housing projects in urban areas, ensuring equitable distribution based on need.

### 10. Maternal Health Services Database

Design a database to track maternal health services in local clinics, including prenatal care, deliveries, and postnatal follow-up.

1. Clean Water Access for Schools Database

Create a database to monitor clean water access in schools, including installation of water points, maintenance schedules, and student health outcomes.

2. Educational Resource Distribution Database

Develop a database to manage the distribution of educational materials (books, laptops, etc.) to schools in underserved regions.

3. Community Microfinance Database

Design a database to track community-based microfinance initiatives, including loans, repayments, and impact on local businesses.

4. Biodiversity Conservation Projects Database

Create a database to manage and monitor conservation projects aimed at protecting endangered species and habitats in Kenya's national parks.

5. Public Health Vaccination Database

Develop a database to track vaccination programs in rural areas, monitoring coverage rates and vaccine availability.

6. Sustainable Agriculture Practices Database

Design a database to promote and track sustainable farming practices among smallholder farmers, focusing on crop diversification and soil conservation.

7. Community Health Worker Management Well- System

Design a database to manage and coordinate community health workers, including training records, assignments, and patient follow-ups.

8. Public Transport Optimization Database

Create a database to track public transport routes, schedules, and usage in urban areas, aiming to optimize efficiency and reduce congestion.

9. Waste Management and Recycling Database

Develop a database to manage waste collection and recycling programs in Kenyan cities, tracking participation rates and recycling outputs.

10. Youth Employment Initiative Database

Design a database to monitor youth employment programs, including job placements, training opportunities, and long-term outcomes.

1. Database for Disaster Resilience

Build a database to manage data on disaster preparedness and response initiatives in vulnerable areas.

2. Tree Planting and Reforestation Database

Develop a database to manage and track tree planting initiatives, including species planted, survival rates, and impact on local ecosystems.

3. Anti-Poaching Effort Management System

Create a database to coordinate anti-poaching efforts in Kenyan wildlife reserves, including patrol schedules, incident reports, and wildlife monitoring.

4. Nutrition and Food Security Monitoring System

Create a database to monitor nutrition programs, tracking food distribution, beneficiary health metrics, and program effectiveness.

5. Maternal and Child Health Database

Develop a database to manage maternal and child health programs, tracking antenatal visits, immunizations, and child growth metrics.

6. Local Tourism Promotion Database

Create a database to promote local tourism in underrepresented regions, tracking visitor numbers, accommodations, and cultural sites.

7. Flood Risk Management Database

Design a database to monitor flood-prone areas, track early warning systems, and coordinate disaster response efforts.

8. Gender-Based Violence Reporting System

Develop a database to track incidents of gender-based violence, including case management, support services, and legal outcomes.

9. Child Education Sponsorship Database

Design a database to manage sponsorship programs for children's education, tracking donations, beneficiary progress, and impact assessments.

10. Access to Justice Database

Design a database to manage legal aid services for underserved communities, tracking cases, outcomes, and client demographics.